AI COMPUTING CONTINUUM



Accelerating the Edge Revolution: AI Inference, vRAN & Telco on AI Native Edge-MHS 1.0

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AI Deployments Landscape

AI Computing Continuum (AICC)

AI Native Edge-MHS Target

DC-MHS → AI-Native Edge-MHS

Location

Cloud Public	Distributed Datacenter (DC) – Private or Public	Access Semi-Private	Edge / On-Prem Private
Hyperscale / Large Enterprise	Small/Med/Large Enterprise, NeoCloud, CoLo	CoSPs, CoLo, PoP	Retail Outlets, Quick Serve, Industrial, Transportation, etc.

Different servers with different management and security

Rack Scale Architecture



AICC Hyperscale Adjacent

4x GPU with no Scale Up
1.5 - 5KW



Sovereign AI
Need for isolation
(no multi-tenancy required)

8x GPU Scale Up
8 - 15KW



Scale Up
(OAC targeted)

1x GPU no Scale Up
0.8KW - 2KW
Short Depth



1x to 4x GPU, no Scale Up
3KW - 5KW
Full Length



Security Critical

Ruggedized
Liquid Cooled



Limited thermal & power delivery capacity

1x GPU no Scale Up
0.8KW - 3KW
Short Depth



The Hyper-scale Adjacent AI Challenges

Democratized hardware operating as a homogeneous system using heterogenous AI hardware in varying deployment infrastructures!

Distributed Deployments Scale Up, Out & Across



Latency & Connectivity



Safety, Privacy & Security



Reliability



Ruggedized & Space Limited



AI-Native Edge-MHS Targets Addressing Some of the Challenges

"Distributed AI must be safe, private and secure; reliable and technician-deployable; real-time connected; and able to run in rugged, space- and power-limited locations."

Safety, Privacy & Security

- Operator-less machinery must avoid in-field accidents
- Sensitive data (soil, rain, crop readiness) stays private
- Hardened against hijacking and malicious attack



Reliability, Maintenance & Deployment

- Deployable by technicians, not engineers
- Lightweight, zero-touch provisioned, resilient
- Rugged NEBS & IP67+ certifiable; remotely managed



Latency & Connectivity

- Nano-second response for high-efficiency lines
- Robot-to-robot and building-to-building AI comms
- Local GPUs + time-synchronized networking via M-XIO



Ruggedized & Space Limited

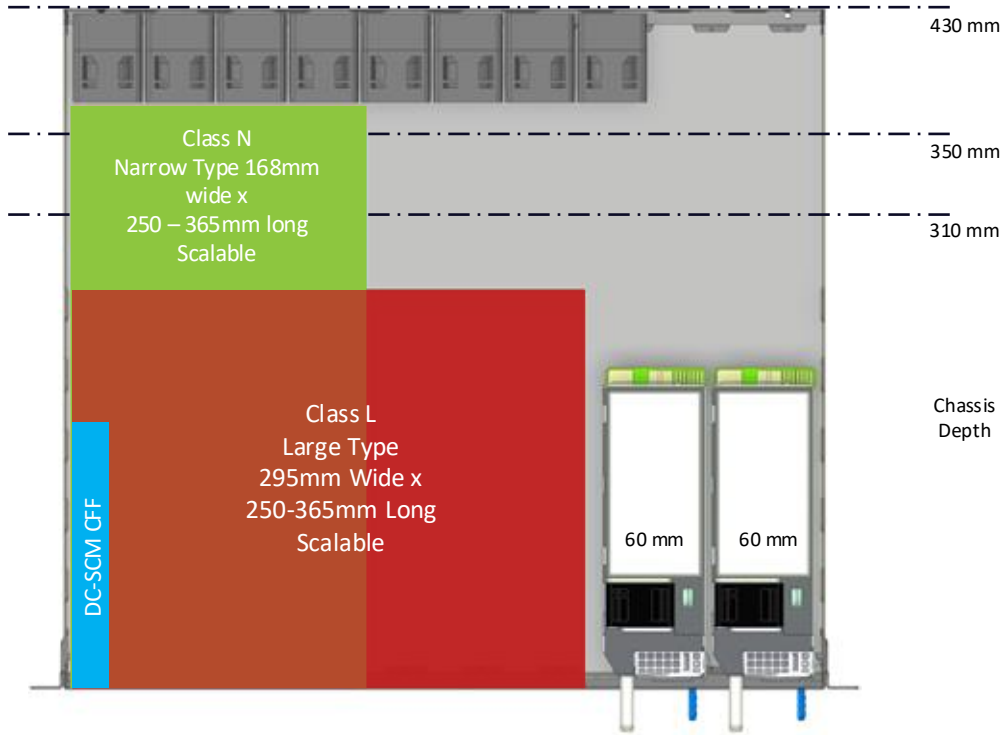
- Legacy cabinets and outdoor, harsh environments
- Varying power sources: 12VAC and -48VDC
- Equipment length < 310mm and < 430mm; limited cooling



But there are more challenges to still solve!

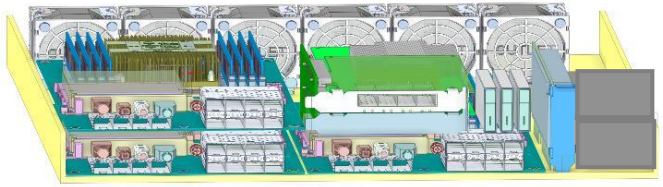
AI-Native Edge-MHS

- Modular hardware system designed for the Edge
- Compatible with DC-MHS where applicable
- Allows system providers to adopt scale more rapidly
- Re-usable in multiple design environments
- A couple of potential board types - 3 distinct problems based on chassis depth: ***Short***, ***Mid***, ***Extended***
- AI Optimized for Edge AI Sleds or AI Add-In Cards
- 19” 1RU or 2RU rack optimized
- Pre-solves technical challenges expected to be faced in Edge solutions
- Timing Module & GNSS Module Support (refer to OCP NIC)
- DC-SCM Compact Form Factor (CFF) Support





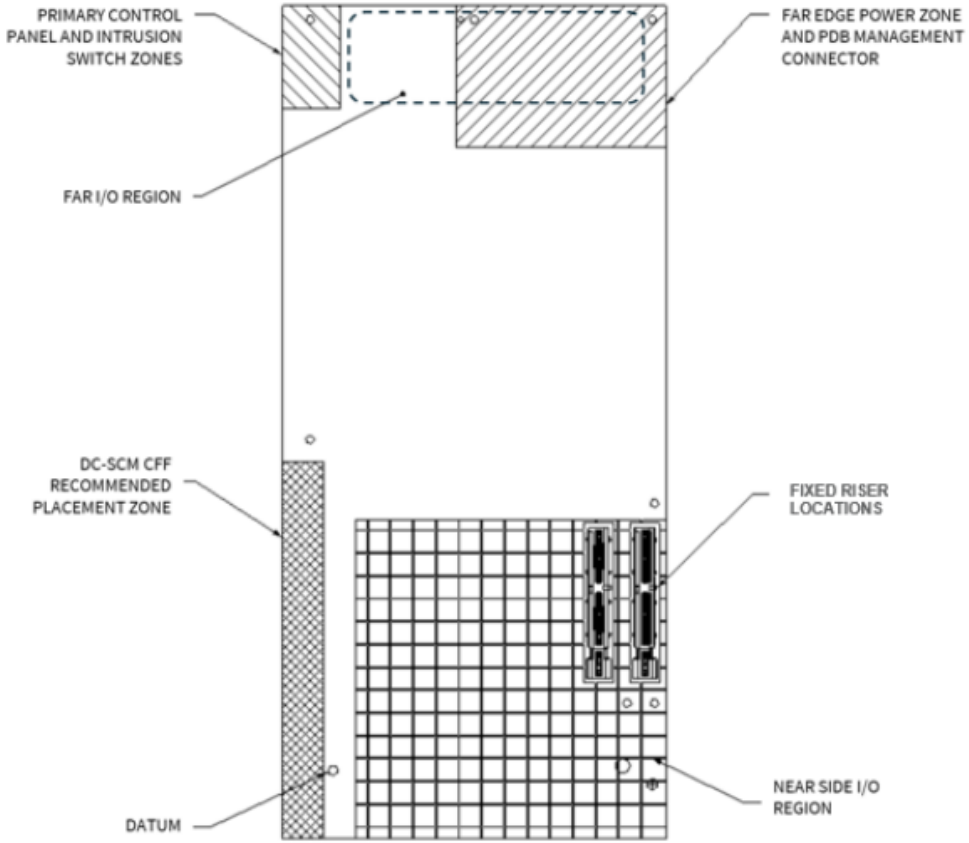
(1U Class L) 310 mm Deep



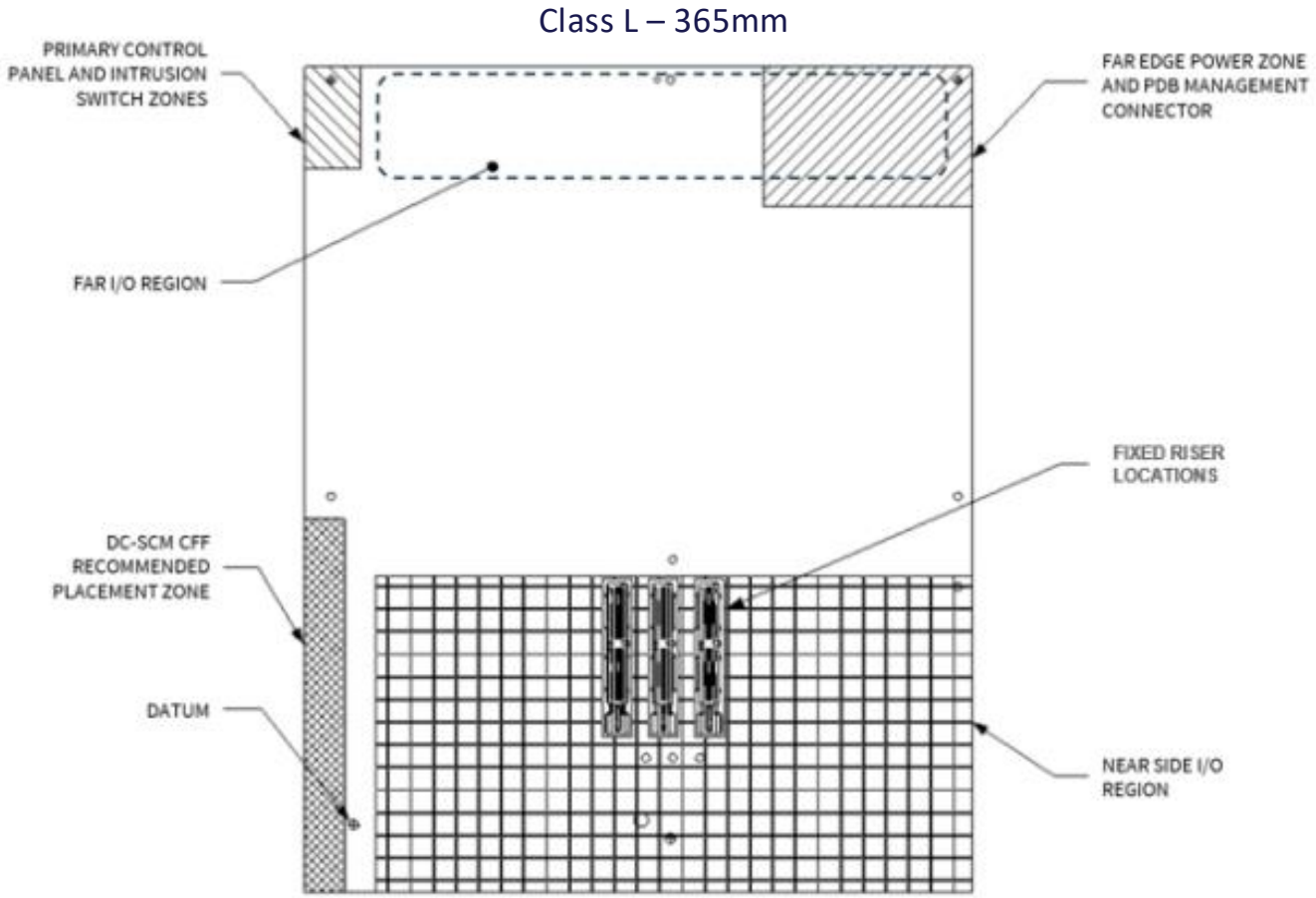
(Two 1U Class N Left, 2U Class N Right) – 430mm Deep

Key Elements of AI Native Edge-MHS

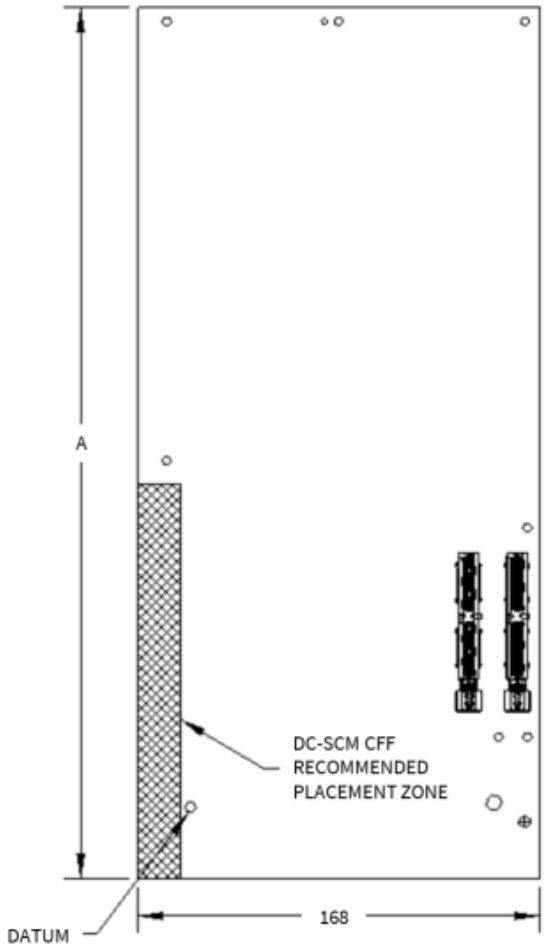
Class N – 365mm



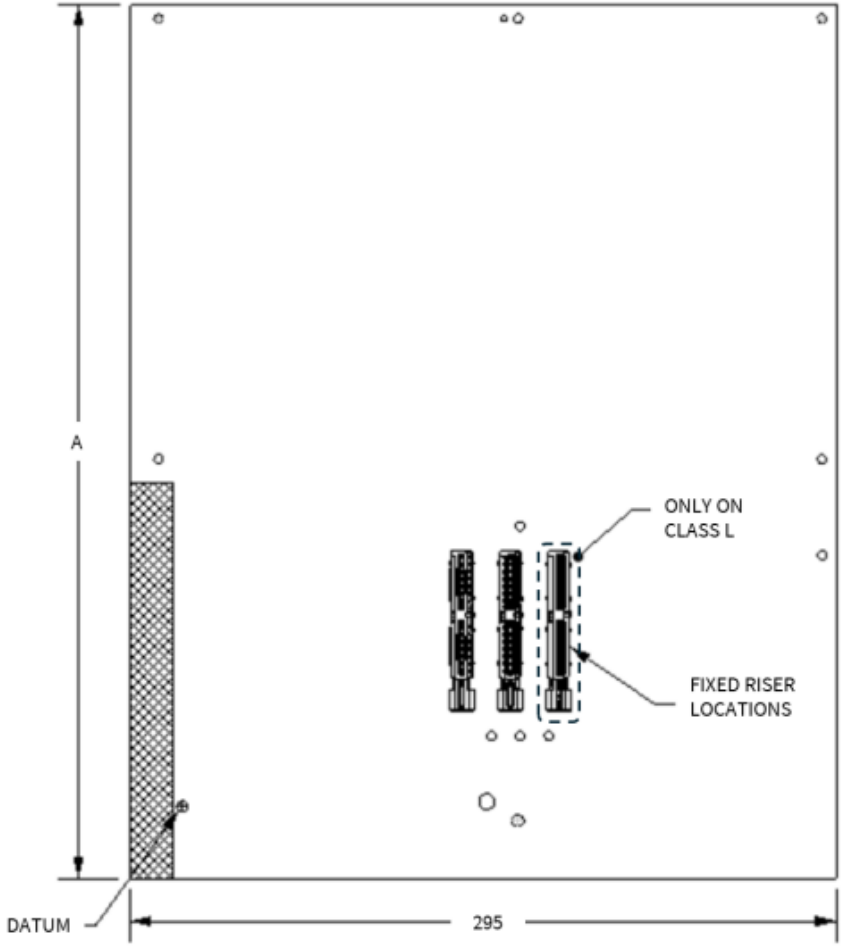
Class L – 365mm



AI NATIVE EDGE-MHS HPM CLASS N AND L OUTLINES



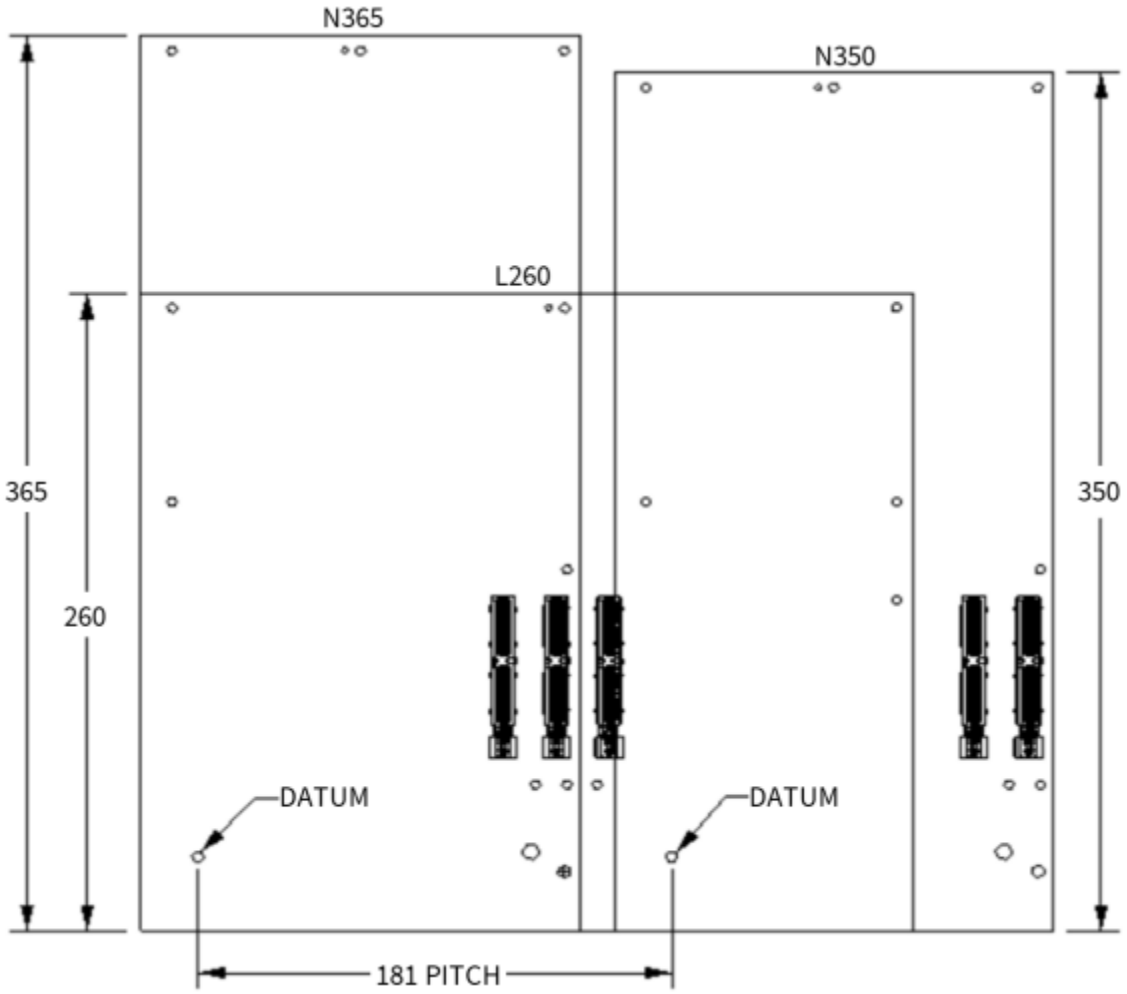
HPM Designator	A	Targeted 19" Server Depth
N260	260mm	310mm Server
N280	280mm	350mm to 420mm Server
N350	350mm	420mm to 450mm Server
N365	365mm	OTII-E Class Server



HPM Designator	A	Targeted 19" Depth
L260	260mm	310mm Deep Server
L305	305.59mm	M-DNO Type 4 Size
L350	350mm	420mm to 450mm Server

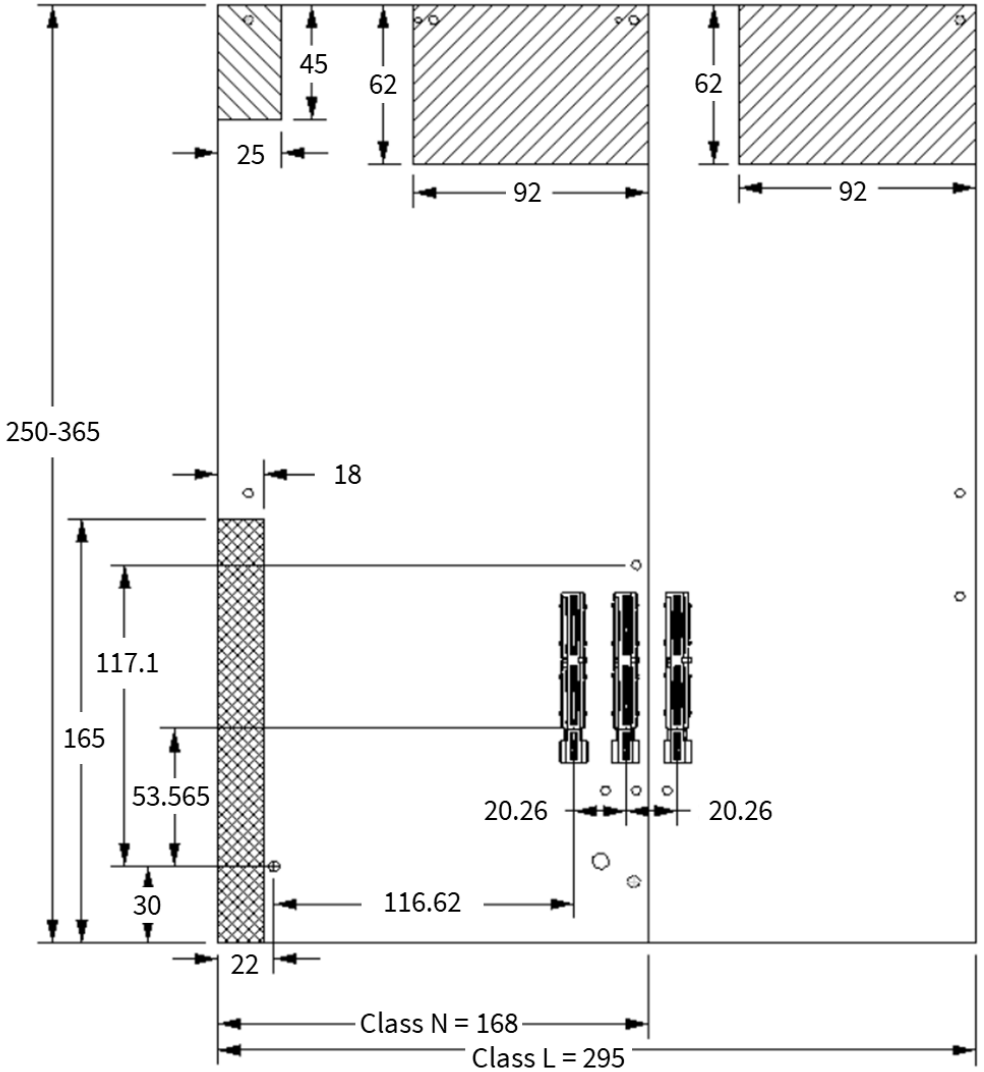
AI NATIVE EDGE-MHS HPM COMPATIBILITY OVERLAYS

- 3 boards shown to the right (Class N365, Class L260, Class N350)
- Common chassis support enabled to support the following configurations:
  - 1 Class N (illustrated by N365 placement)
  - 1 Class L (illustrated by L260 placement)
  - 2 Class N (illustrated by the N365 and N350 placement)
- Allows system providers to adopt different solutions more easily
- Allows chassis providers to develop common chassis support
- Allows common mounting holes and common riser assemblies across multiple configurations
- When implementing 181-millimeter pitch between N365 and N350 to allow mounting points to work when trying to utilize common mounting holes and risers with a the L260 solution

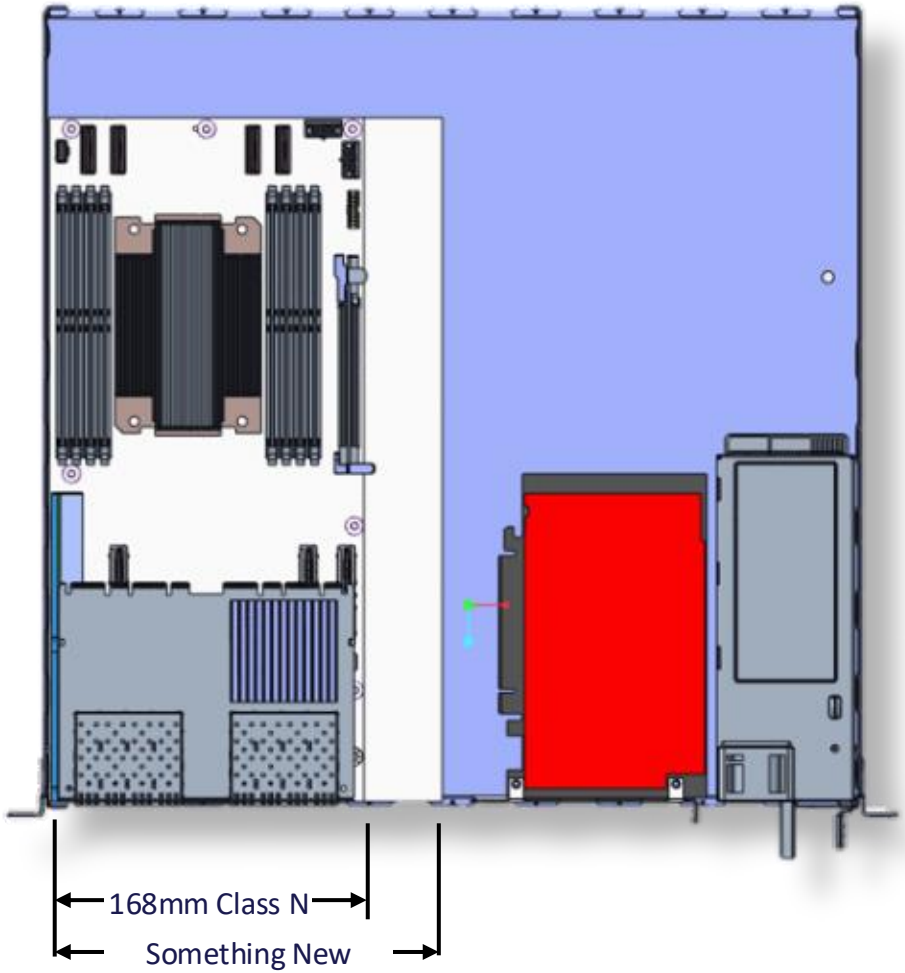


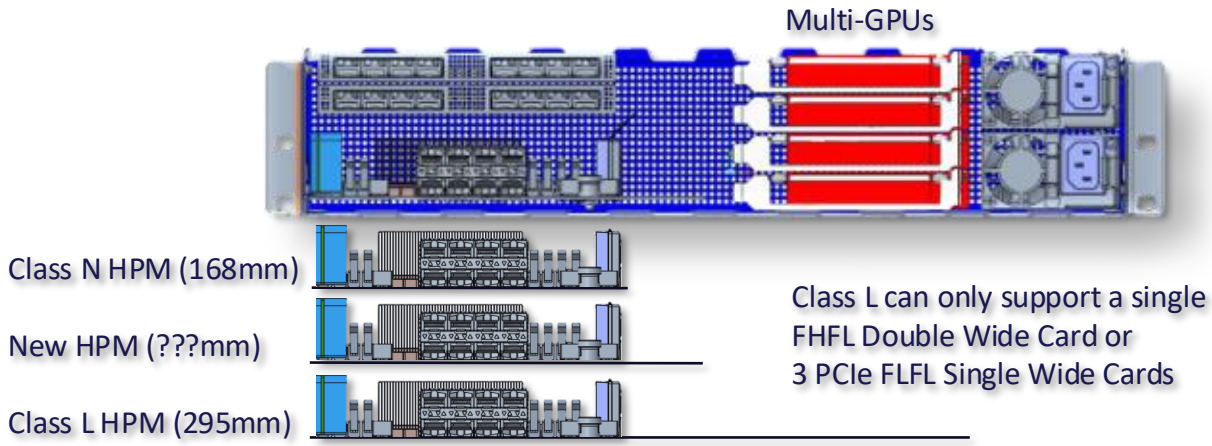
AI NATIVE EDGE-MHS HPM KEY DIMENSIONS

- A Class N (168mm wide) and Class L (295mm wide) are shown in a single diagram to illustrate how one is an extension of the other
- HPM's are scalable in length between 250-365mm
- The control panel and intrusion switch zone is 25mm x 45 mm and always resides in the upper left corner of the HPM
- The power and PDM management connector is located in the upper right corner. It intentionally extends to over halfway past the middle of the Class N HPM to allow blind mate solutions to be compliant to the specification
- Allows common riser mounting locations and positions between a Class N and Class L solution.
- Recommended zone for DC-SCM CFF is 18mm x 22mm but is not required for the solution to be compatible (note if a BMC down solution is implemented it must be SW compatible to a DC-SCM CFF to be considered compliant to the specification)



AI Native MHS Future Direction – New HPM Size





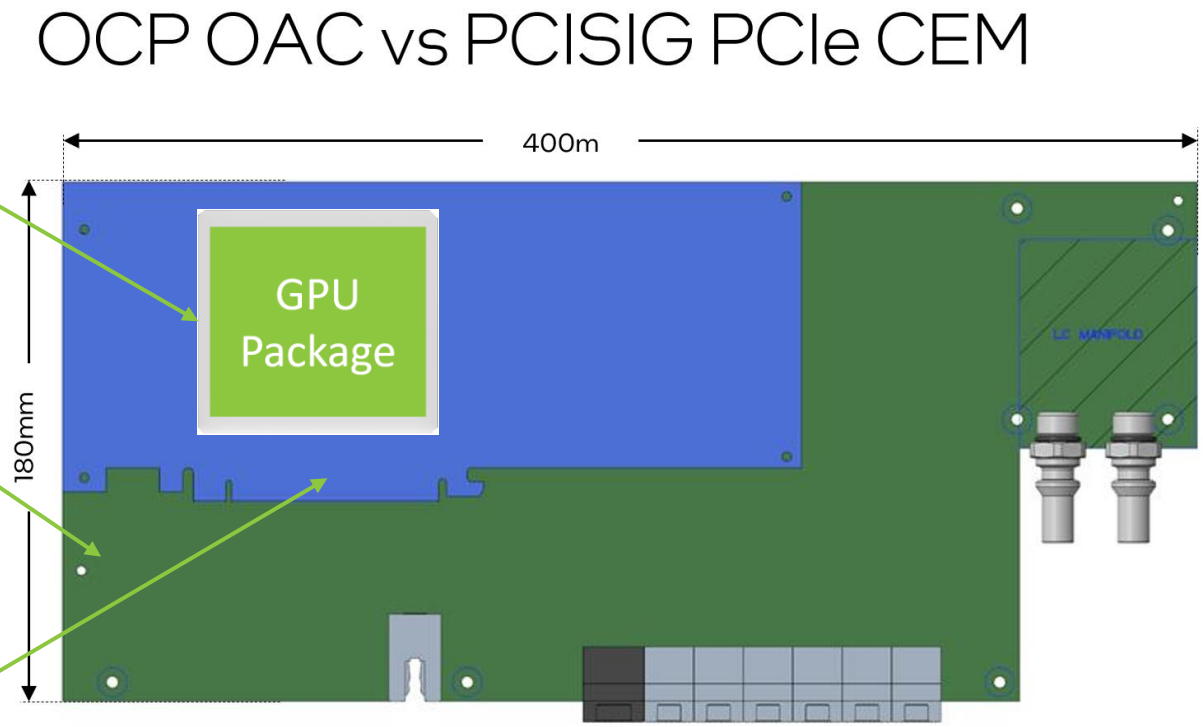
Memory and I/O densification coupled with the need for GPUs are driving AI Native MHS to explore new form factor between Class N and L

AI Native MHS Future Direction – OAC, OCP NIC, and PCIe Good Enough?

GPU and Switch sizes are large

OAC is very large for 1U and 2U systems but very complete functionally

PCIe CEM offers no native coherent (Scale-Up) link connectivity other than I/O Panel



- Scale is much different, up to 120KW type racks (mostly below 50KW) and 600W Air-cooled GPU cards maximum
- 19” and 21” deployments allowing different types of scale due to legacy deployments
- GPUs are up to 1000W/GPU versus 5000W in hyperscale but below 500W in hyperscale adjacent environments
- Design optimized for air but can be liquid cooled

Are current solutions sufficient or is there something else needed?

Call to Action

- Compute and AI needs beyond traditional datacenter walls are accelerating
  - Distributed AI solutions are becoming a reality but are not without a challenge
- The AICC leveraging AI Native Edge-MHS is designed with these challenges in mind
- AI-Native Edge-MHS Specification V1.0 Released
- Additional information available on the AI Native Edge-MHS Wiki
- https://www.opencompute.org/w/index.php?title=AICC/AI_Native_Edge_MHS
- Provide feedback (Workstream Leads)
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Thank You!

